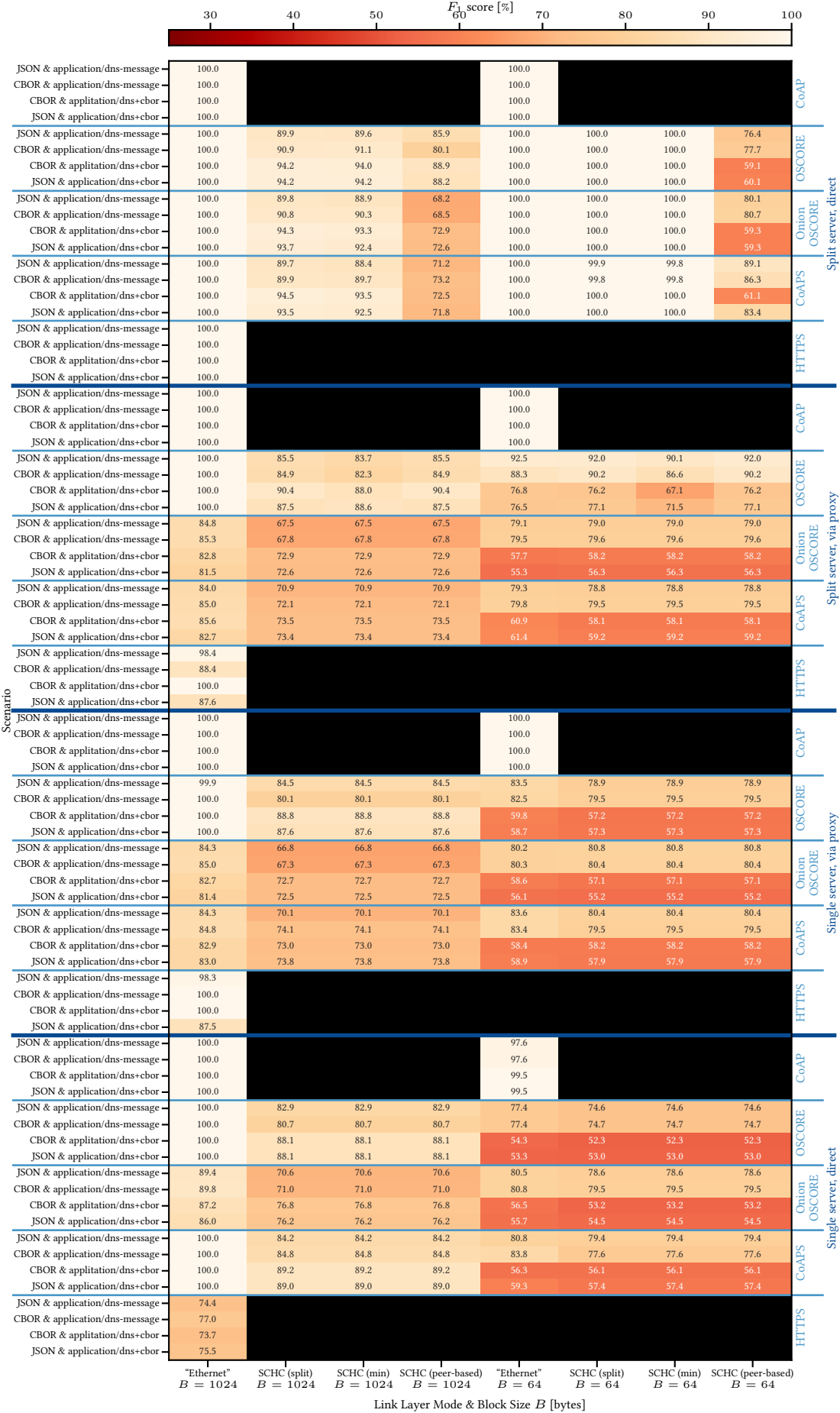
F_1 score [%]
30
40
50
60
70
80
90
100
Scenario
Link Layer Mode & Block Size B [bytes]
"Ethernet" $B = 1024$
SCHC (split) $B = 1024$
SCHC (min) $B = 1024$
SCHC (peer-based) $B = 1024$
"Ethernet" $B = 64$
SCHC (split) $B = 64$
SCHC (min) $B = 64$
SCHC (peer-based) $B = 64$
Split server, direct
JSON & application/dns-message 100.0 100.0
CBOR & application/dns-message 100.0 100.0
CBOR & applitation/dns+cbor 100.0 100.0
JSON & application/dns+cbor 100.0 100.0
CoAP
JSON & application/dns-message 100.0 89.9 89.6 85.9 100.0 100.0 100.0 76.4
CBOR & application/dns-message 100.0 90.9 91.1 80.1 100.0 100.0 100.0 77.7
CBOR & applitation/dns+cbor 100.0 94.2 94.0 88.9 100.0 100.0 100.0 59.1
JSON & application/dns+cbor 100.0 94.2 94.2 88.2 100.0 100.0 100.0 60.1
OSCORE
JSON & application/dns-message 100.0 89.8 88.9 68.2 100.0 100.0 100.0 80.1
CBOR & application/dns-message 100.0 90.8 90.3 68.5 100.0 100.0 100.0 80.7
CBOR & applitation/dns+cbor 100.0 94.3 93.3 72.9 100.0 100.0 100.0 59.3
JSON & application/dns+cbor 100.0 93.7 92.4 72.6 100.0 100.0 100.0 59.3
Onion OSCORE
JSON & application/dns-message 100.0 89.7 88.4 71.2 100.0 99.9 99.8 89.1
CBOR & application/dns-message 100.0 89.9 89.7 73.2 100.0 99.8 99.8 86.3
CBOR & applitation/dns+cbor 100.0 94.5 93.5 72.5 100.0 100.0 100.0 61.1
JSON & application/dns+cbor 100.0 93.5 92.5 71.8 100.0 100.0 100.0 83.4
CoAPS
JSON & application/dns-message 100.0
CBOR & application/dns-message 100.0
CBOR & applitation/dns+cbor 100.0
JSON & application/dns+cbor 100.0
HTTPS
Split server, via proxy
JSON & application/dns-message 100.0 100.0
CBOR & application/dns-message 100.0 100.0
CBOR & applitation/dns+cbor 100.0 100.0
JSON & application/dns+cbor 100.0 100.0
CoAP
JSON & application/dns-message 100.0 85.5 83.7 85.5 92.5 92.0 90.1 92.0
CBOR & application/dns-message 100.0 84.9 82.3 84.9 88.3 90.2 86.6 90.2
CBOR & applitation/dns+cbor 100.0 90.4 88.0 90.4 76.8 76.2 67.1 76.2
JSON & application/dns+cbor 100.0 87.5 88.6 87.5 76.5 77.1 71.5 77.1
OSCORE
JSON & application/dns-message 84.8 67.5 67.5 67.5 79.1 79.0 79.0 79.0
CBOR & application/dns-message 85.3 67.8 67.8 67.8 79.5 79.6 79.6 79.6
CBOR & applitation/dns+cbor 82.8 72.9 72.9 72.9 57.7 58.2 58.2 58.2
JSON & application/dns+cbor 81.5 72.6 72.6 72.6 55.3 56.3 56.3 56.3
Onion OSCORE
JSON & application/dns-message 84.0 70.9 70.9 70.9 79.3 78.8 78.8 78.8
CBOR & application/dns-message 85.0 72.1 72.1 72.1 79.8 79.5 79.5 79.5
CBOR & applitation/dns+cbor 85.6 73.5 73.5 73.5 60.9 58.1 58.1 58.1
JSON & application/dns+cbor 82.7 73.4 73.4 73.4 61.4 59.2 59.2 59.2
CoAPS
JSON & application/dns-message 98.4
CBOR & application/dns-message 88.4
CBOR & applitation/dns+cbor 100.0
JSON & application/dns+cbor 87.6
HTTPS
Single server, via proxy
JSON & application/dns-message 100.0 100.0
CBOR & application/dns-message 100.0 100.0
CBOR & applitation/dns+cbor 100.0 100.0
JSON & application/dns+cbor 100.0 100.0
CoAP
JSON & application/dns-message 99.9 84.5 84.5 84.5 83.5 78.9 78.9 78.9
CBOR & application/dns-message 100.0 80.1 80.1 80.1 82.5 79.5 79.5 79.5
CBOR & applitation/dns+cbor 100.0 88.8 88.8 88.8 59.8 57.2 57.2 57.2
JSON & application/dns+cbor 100.0 87.6 87.6 87.6 58.7 57.3 57.3 57.3
OSCORE
JSON & application/dns-message 84.3 66.8 66.8 66.8 80.2 80.8 80.8 80.8
CBOR & application/dns-message 85.0 67.3 67.3 67.3 80.3 80.4 80.4 80.4
CBOR & applitation/dns+cbor 82.7 72.7 72.7 72.7 58.6 57.1 57.1 57.1
JSON & application/dns+cbor 81.4 72.5 72.5 72.5 56.1 55.2 55.2 55.2
Onion OSCORE
JSON & application/dns-message 84.3 70.1 70.1 70.1 83.6 80.4 80.4 80.4
CBOR & application/dns-message 84.8 74.1 74.1 74.1 83.4 79.5 79.5 79.5
CBOR & applitation/dns+cbor 82.9 73.0 73.0 73.0 58.4 58.2 58.2 58.2
JSON & application/dns+cbor 83.0 73.8 73.8 73.8 58.9 57.9 57.9 57.9
CoAPS
JSON & application/dns-message 98.3
CBOR & application/dns-message 100.0
CBOR & applitation/dns+cbor 100.0
JSON & application/dns+cbor 87.5
HTTPS
Single server, direct
JSON & application/dns-message 100.0 97.6
CBOR & application/dns-message 100.0 97.6
CBOR & applitation/dns+cbor 100.0 99.5
JSON & application/dns+cbor 100.0 99.5
CoAP
JSON & application/dns-message 100.0 82.9 82.9 82.9 77.4 74.6 74.6 74.6
CBOR & application/dns-message 100.0 80.7 80.7 80.7 77.4 74.7 74.7 74.7
CBOR & applitation/dns+cbor 100.0 88.1 88.1 88.1 54.3 52.3 52.3 52.3
JSON & application/dns+cbor 100.0 88.1 88.1 88.1 53.3 53.0 53.0 53.0
OSCORE
JSON & application/dns-message 89.4 70.6 70.6 70.6 80.5 78.6 78.6 78.6
CBOR & application/dns-message 89.8 71.0 71.0 71.0 80.8 79.5 79.5 79.5
CBOR & applitation/dns+cbor 87.2 76.8 76.8 76.8 56.5 53.2 53.2 53.2
JSON & application/dns+cbor 86.0 76.2 76.2 76.2 55.7 54.5 54.5 54.5
Onion OSCORE
JSON & application/dns-message 100.0 84.2 84.2 84.2 80.8 79.4 79.4 79.4
CBOR & application/dns-message 100.0 84.8 84.8 84.8 83.8 77.6 77.6 77.6
CBOR & applitation/dns+cbor 100.0 89.2 89.2 89.2 56.3 56.1 56.1 56.1
JSON & application/dns+cbor 100.0 89.0 89.0 89.0 59.3 57.4 57.4 57.4
CoAPS
JSON & application/dns-message 74.4
CBOR & application/dns-message 77.0
CBOR & applitation/dns+cbor 73.7
JSON & application/dns+cbor 75.5
HTTPS